College & Career ROI — Cheat Sheet

Financial Algebra · Unit 9: College & Career ROI

Essential vocabulary

Cost of attendance (COA)
The total estimated yearly cost of college: tuition, fees, housing, food, books, transportation, and personal expenses

Sticker price
The published cost of attendance before any financial aid

Net price
Cost of attendance minus grants and scholarships — what the family must actually pay or borrow

Grant
Financial aid, usually need-based, that does not have to be repaid

Scholarship
Gift aid, often merit-based, that does not have to be repaid

Work-study
A financial aid program that provides part-time jobs to help students earn money for college costs

Subsidized loan
A federal student loan for which the government pays the interest while the student is enrolled at least half-time

Unsubsidized loan
A student loan that accrues interest from the time it is disbursed, including during school

Capitalized interest
Unpaid interest that is added to the loan principal, so future interest is charged on it

FAFSA
The Free Application for Federal Student Aid, used to determine eligibility for federal grants, loans, and work-study

Return on investment (ROI)
The gain from an investment compared with its cost, often expressed as a percent or a payback period

Key formulas

Cost of attendance	tuition + fees + housing + food + books + other
Net price	COA − grants − scholarships
Loan payment	$\text{PMT} = \frac{P \cdot i}{1 - (1 + i)^{(-n)}}$
Capitalized balance	principal + unpaid interest
Earnings with raises	$S = \frac{a((1 + g)^n - 1)}{g}$
Payback period	$\frac{\text{total cost}}{\text{yearly earnings gain}}$
Debt-to-income (loans)	$\frac{\text{yearly loan payments}}{\text{gross income}}$

Types of aid

Gift	Grants, scholarships — no repayment
Earned	Work-study
Borrowed (student)	Subsidized, unsubsidized
Borrowed (parent)	Parent PLUS

Payback period
The time needed for extra earnings to cover the cost of an investment

Opportunity cost of education
Earnings given up while in school instead of working full-time

Apprenticeship
A paid training program combining on-the-job learning with instruction, leading to a credential

Lifetime earnings
The total income earned over a working career

Completion rate
The share of students who finish a program within a set time

Common misconceptions

- Loans reduce the price.
- College is always/never worth it.
- Sticker price = what you pay.
- Starting salary is the only number that matters.

Worked example

1. COA $30,500 − grants $13,000 = $17,500 net
2. Borrow $5,500 × 4 = $22,000 at 6.5%, 10 yr
3. Payment ≈ $249.81/month
4. Starting salary $55,000 → payments ≈ 5.5% of gross

Questions to ask before deciding

- What is the net price for me?
- How much will I borrow in total?
- What do graduates of this program earn?
- What is the completion rate?
- What are my alternatives (transfer, apprenticeship, certificate)?

Decision tips

- File the FAFSA every year.
- Compare net prices, not sticker prices.
- Borrow no more than your expected first-year salary.
- Pay interest during school if you can.

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